



BENCH POWER SUPPLY

Using A Computer's Power Supply

JAYWARDHAN PANDIT

A good bench power supply is the most vital equipment for electronics engineers and hobbyists. However, bench power supplies used in R&D labs are costly. Output rating is also a major issue. A variable DC supply up to 30V with a few amperes current is the basic requirement for labs.

Making a power supply for use in labs is not an easy task due to the complexity of the circuit. To provide high current outputs, it needs heavy transformers and semiconductor

devices. A simple alternative is to use a computer power supply unit (PSU). With some modifications, the PSU can be turned into a good bench power supply.

Fig. 1 shows input and output ratings of an ATX computer PSU. The available outputs are +3.3V (20A), +5V (34A), +12V (16A), -5V (0.5A), -12V (0.5A) and +5VSB (2A).

These outputs are available on one 20-pin Molex connector (some newer PSUs have 24-pin connector) and four various other 4-pin connectors.

A 20-pin Molex connector is shown in Fig. 2. Wire colour codes of standard ATX PSU connector are shown in the table below. From the table, you can identify voltage outputs corresponding to different colour codes.

Power up and testing

To test a computer PSU without connecting it to the motherboard, short its green wire to any of the black wires (ground wires). The power supply's cooling fan should turn on to indicate that the power supply is 'on.' Now you can check the volt-

ages of all the different wires with respect to black wire by using a multimeter. If all the voltages are as per the colour code table, the power supply is working fine.

Another method is to check the voltage of grey wire, which is a PGS or power-good signal. It should be +5V approx.

Opening the power supply unit

As you know, computers work on 230V AC power. To open the PSU enclosure, first disconnect its 230V supply cord. After you unscrew the PSU enclosure and remove the cover, be careful of large capacitors on the PCB. Wait for a while or discharge them using a proper discharging tool. Otherwise, you run the risk of getting an electric shock from capacitors.

You will see that all the same-colour wires are bundled together and soldered in one place on the PCB. The PCB is also marked with legends for their voltage and signals.

Modifications

Fixed outputs. Since the outputs available on various PSU connectors



Fig. 1: Input and output ratings of an ATX computer PSU



Fig. 2: 20-pin ATX Molex connector

COLOUR CODES OF PSU WIRES

Colour code	Voltage/Signal	Description
Yellow	12V	+12V DC
Red	5V	+5V DC
Orange	3.3V	+3.3V DC
Green	PS_ON	Power Supply On (active-low)
Blue	-12V	-12V DC
White	-5V	-5V DC
Purple	+5VSB	+5V DC standby voltage
Grey	PGS (+5V)	Power-good signal (+5V DC when outputs are ok)
Black	Common	Ground